

Gingerbread's Grand Gables



A 6×6 all-ways high-volatility Christmas slot — a deliberately starved base game carried by two bounded, mathematically-tuned bonus features.

GAME RTP

~96%

VOLATILITY

High

MAX WIN

35,070×

GRID

6×6 All-Ways

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Original concept, math design, Monte-Carlo validation, art direction & engineering. All figures sourced from the game's real configuration and simulation harness.
Confidential design document.

1 Overview

Gingerbread's Grand Gables is an original 6×6 all-ways Christmas slot built in a glossy 2.5D hyper-casual style. Wins pay left-to-right from the leftmost reel across matching symbols, with no fixed paylines and no base-game cascade. The design philosophy is taken from modern high-volatility titles such as *Wanted Dead or a Wild*: the base game is intentionally lean, and the return-to-player is carried almost entirely by two contrasting bonus features — one frequent and steady, one rare and explosive.

THE HOOK

Two bonuses, two volatility profiles. **Raining Houses** (cheap, frequent, capped) gives steady action; **Build the Gables** (premium, rare, barbell-shaped) chases a 35,000×+ ceiling. Each is independently tuned to ~96% of its cost, and the base-game scatter weights are tuned so the whole game lands at ~96% RTP through natural play.

ATTRIBUTE	VALUE
Title	Gingerbread's Grand Gables
Grid / mechanic	6×6, all-ways (left-to-right, 3+ from reel 1), no base cascade
Paying symbols	2 premiums, 3 mids, 4 lows, wild, 2 scatters
Theme / style	Christmas gingerbread · glossy 2.5D hyper-casual
Base game RTP (line only)	~36.8%
Total game RTP (with natural bonus triggers)	~96%
Volatility class	High
Max win	35,070× (validated ceiling)
Engine / stack	TypeScript · PixiJS v8 · GSAP · Vite

2 Key Stats

TOTAL GAME RTP

~96%

Base line + natural bonus triggers

BUILD BONUS RTP

96.04%

of 350× cost · 1M-spin sim

RAINING RTP

98.41%

of 100× cost · 200k-spin sim

MAX WIN

35,070×

Ultra Palace × max multiplier

METRIC	VALUE	VALIDATION
Total game RTP	~96%	Monte-Carlo over real game logic, 1M base spins
Base game line RTP	~36.8%	1M-spin sim
Build the Gables bonus RTP	96.04%	1,000,000-bonus sim
Raining Houses bonus RTP	98.41%	200,000-bonus sim
Build the Gables trigger rate (natural)	~1 in 2,294	scatter-weight sim
Raining Houses trigger rate (natural)	~1 in 215	scatter-weight sim
Build bonus 0× (dud) rate	0.00%	multiplier floor = 1
Volatility class	High	barbell payout distribution

3 Core Mechanics

Base Spin & Win Evaluation

Each spin fills the 6×6 grid from weighted reels. Evaluation is **all-ways**: a win is any symbol appearing on 3+ consecutive reels starting from reel 1, with the win value equal to the symbol's pay multiplied by the number of "ways" (the product of that symbol's count on each contributing reel). There is **no base-game cascade** — each spin is evaluated once. Minimum match lengths differ by tier: premiums pay from 4 of a kind, mids and lows from 5.

Paytable & Symbol Weights

SYMBOL	TIER	MIN	REEL WEIGHT
Gingerbread Man	Premium	4+	[VERIFY]
Gingerbread Woman	Premium	4+	[VERIFY]
Gumdrop	Mid	5+	[VERIFY]
Candy Cane	Mid	5+	[VERIFY]
Tree Cookie	Mid	5+	[VERIFY]
Ten / Jack / Queen / King	Low	5+	[VERIFY]
Wild	Special	—	3 [VERIFY]
Scatter — Build (blueprint)	Trigger	3+	~2
Scatter — Rain (falling house)	Trigger	3+	~6

Per-symbol pay values by match length are defined in `paytable.ts` and require sourcing from the live file — see Verify section.

Wilds & Scatters

The **wild** substitutes for all paying symbols (substitute-only; it does not pay a line itself). Two distinct **scatter** symbols each trigger their own bonus on 3+: the blueprint scatter triggers Build the Gables, the falling-house scatter triggers Raining Houses. Only one scatter type is eligible per base spin, removing mixed-trigger ambiguity. If both were to qualify, Build the Gables takes priority.

DESIGN NOTE — TWO-SCATTER TRIGGER

Rather than a single scatter opening a choice popup (which would let players always take the more valuable bonus for free), each bonus has its own scatter with its own weight. The premium bonus's scatter is far rarer, which both protects the value of its 350× buy and keeps the natural-trigger RTP contribution controllable.

4 Bonus Features

Build the Gables — Premium Bonus (buy 350×)

A two-phase hoard-then-showdown feature. **Phase 1 (Collect)**: a respin board where bricks accumulate. Each respin can drop a brick (1.40% per cell), an icing +1 (0.40%) or icing +2 (0.10%); collecting anything resets respins to 3, three dead respins end the phase, and the board caps at 36 bricks. The brick total resolves to a house tier; the multiplier starts at **1** and climbs with collected icing. **Phase 2 (Showdown)**: theatrical reveal spins that pay a fixed tier prize × the collected multiplier — bounded by design, never an unbounded all-ways evaluation.

TIER	BRICKS	PRIZE (×BET)	FREQUENCY
Cottage	4	2.65	40.56%
Manor	8	8.00	32.11%
Mansion	14	18.50	17.64%
Palace	22	45.00	5.20%
Grand Palace	28	93.00	2.89%
Ultra Palace	36	1002.00	1.59%

Frequencies from the 1M-bonus validation run. Cottage is a guaranteed floor (bricks < 4 still resolve to Cottage). Final payout = TIER_PRIZE × multiplier × bet.

VALIDATED — BUILD THE GABLES

RTP **96.04%** of 350× · max win **35,070×** · **0%** dud rate · barbell distribution: ~64% of bonuses pay under 50×, ~3.6% pay over 1000×. Ultra Palace at a stacked multiplier produces the game's ceiling.

Raining Houses — Standard Bonus (buy 100×)

12 free spins. Each spin, one outcome is drawn from a weighted table: ~68.5% **no house**, otherwise a gingerbread house of one of seven sizes (1×1 up to a full-board 6×6 Ultra Palace) rains onto a valid position and acts as a **non-sticky wild** for that spin only. The win evaluates all-ways with that house as wild, then is bounded by a **per-spin cap of 400×**. A 3% chance per spin adds a ×2/×3 icing multiplier; 3+ scatters retrigger +5 spins.

OUTCOME	WEIGHT	NOTE
No house	68.5	dry spin — bounds RTP, adds suspense
1×1 single wild	17.90	common
Cottage 2×2	10.24	
Manor 2×4	2.89	
Mansion 4×4	0.315	
Palace 4×6	0.089	rare
Grand 5×6	0.049	rare
Ultra 6×6	0.020	full-board wild · ~1 in 5,000 spins

VALIDATED — RAINING HOUSES

RTP **98.41%** of 100× · max win **1,532×** · cap fires on ~1.04% of spins. Steadier, lower-ceiling profile — the deliberate counterweight to Build the Gables.

5 Math & RTP Design

The whole game was tuned against a **Monte-Carlo harness that runs the real game logic** — not an abstract spreadsheet model. The simulator mirrors the engine's RNG, weights, and payout code exactly; whenever the two diverged, the game was treated as source of truth and the sim was corrected to match, then re-tuned. Every headline number below comes from that harness.

Target & Result

Target: ~96% total RTP, high volatility, with the base game starved and the bonuses carrying the return. Final validated result: base line ~36.8%, Raining Houses contributing ~45.8% ($1/215 \times 98.4\times$), Build the Gables ~14.7% ($1/2294 \times 336\times$), for a total of ~96% through natural play.

```
// total base-game RTP — how the ~96% is composed
TOTAL = base_line(36.8%) + rain_trigger_rate(1/215) × rain_avg(98.4×) // ≈ 45.8%
+ build_trigger_rate(1/2294) × build_avg(336×) // ≈ 14.7%
// ≈ 96%
```

Key Tuning Levers (in order of leverage)

LEVER	EFFECT
Build Phase-1 brick rate	Controls the tier distribution. Initially produced Ultra Palace on 41% of bonuses; lowered to 1.40% to make Ultra a ~1.6% rare event.
Tier prize table	Sets Build RTP once the distribution is fixed; tuned down ~8% after the multiplier-floor fix.
Build scatter weight	Highest-leverage base-game dial: each natural Build trigger is worth ~336×, so small weight changes move total RTP fast.
Raining no-house % + per-spin cap	Bounds Raining's average; no-house tuned to 68.5%, cap to 400×.

Problems Solved During Tuning

THE EXPLOSION

Both bonuses were first built to pay all-ways over a board of accumulated wilds. With a full-board wild, all-ways evaluation compounds exponentially — RTP readings ran into the **millions of percent**. The fix was structural: replace all-ways-over-wilds with **bounded payout models** — a fixed tier-prize table for Build, a per-spin cap for Raining.

THE FEEL-BUG

Build's multiplier originally started at 0, so collecting no icing meant a built house paid **nothing** — a 9.5% "won a bonus, got zero" rate that the average RTP hid. Flooring the multiplier at 1 removed the dud rate entirely (0%); the tier prizes were then re-tuned down to hold 96%.

THE IMPOSSIBLE TARGET

An early lush board-frame target proved mathematically un-fittable (measured via alpha-channel analysis: its clear opening forced the frame 409px taller than the screen allowed). Documented here as an example of measure-first problem-solving — the board was rescaled to create the needed margin rather than distorting the asset.

6 Design Intent

The intended experience is a lean, suspenseful base game punctuated by two distinct bonus fantasies. Most base spins are quiet by design (high dead-spin rate), which makes a scatter trigger feel like an event. The two bonuses deliberately occupy opposite ends of the volatility spectrum so the player has a genuine choice of fantasy.

Raining Houses — the steady play

Frequent, cheaper, capped at 1,532×. A run of dry spins broken by a house raining in; the rare full-board Ultra is the spike. Low-variance comfort.

Build the Gables — the dream chase

Premium, rare, barbell-shaped. Most buys disappoint (~64% under 50×); the return lives in rare Ultra Palace hits at stacked multipliers, up to 35,070×. This is the high-roller fantasy.

VOLATILITY AS A DELIBERATE LEVER

Both bonuses sit at the same ~96% RTP but feel completely different. The 23× gap between their ceilings (35,070× vs 1,532×) is the design statement: identical fairness, opposite experience. The "feels punishing" of Build is the volatility working as intended, not a flaw.

7 Tech & Architecture

AREA	IMPLEMENTATION
Engine / stack	TypeScript (strict) · PixiJS v8 · GSAP · Vite
Math model	Server-authoritative: a dev MockServer owns all RNG & outcomes; the client only renders
Single source of truth	All weights, prizes, thresholds & caps live in config; the sim mirrors them exactly
Validation	Python Monte-Carlo harness running the real game logic (50k tuning runs, 200k–1M confirmation runs)
Info screen	Data-driven payable overlay — reads live from the real constants, can never drift from the math
Audio	Procedurally synthesized SFX (numpy/scipy) + state-driven background music per game mode

⚠ Verify Before Publishing

Figures below were assembled from this project's established validated values. Confirm each against the live code / sim before publishing — these were not re-audited from source files in this build.

[VERIFY] Per-symbol pay values & reel weights — source the exact numbers from `paytable.ts` and the reel-weight config (premiums, mids, lows, wild weight). The payable table lists tiers & min-lengths confirmed, but per-symbol pay multipliers and weights are placeholders.

[VERIFY] Bet range / coin configuration — confirm min/max bet and bet steps from config; not stated in this document.

[VERIFY] Base line RTP (~36.8%) & hit frequency — confirm the exact base-game line RTP and base-game hit frequency / dead-spin % from the latest sim output.

[VERIFY] Total game RTP (~96%) — the 5M confirmation run was requested; confirm the final settled total-RTP figure and its sample size.

[VERIFY] Scatter weights (Build ~2 / Rain ~6) — confirm the final tuned scatter weights and resulting trigger rates from the live MockServer.

[VERIFY] Ultra Palace prize (1002×) & max win (35,070×) — both confirmed in-session at 1M; re-confirm against current HouseTierManager and the latest max-win readout.

[VERIFY] Stale comments — Board.ts contained stale geometry comments (570/170) superseded by constants (540/75). Audit for any other stale comments before quoting code.